

I am a research engineer and a prospective graduate student; my research interests lie in the intersection of AI, Psychology, Cognitive Sciences, and Human-Computer Interaction. My research focuses on building context-aware, multimodal AI-driven systems that can sense and adapt to users' behavioral and cognitive states; in particular, I want to develop cognitive augmentation tools that can effectively support users' learning and memory, in the domain of education.

EDUCATION & ACADEMIC QUALIFICATIONS

Bachelor of Computing in Computer Science (with Honours) , <i>National University of Singapore</i>	Aug 2020 — Jun 2025
Fall Exchange Program , <i>Seoul National University</i>	Sep 2022 — Feb 2023

PUBLICATIONS & ONGOING WORK

Don't Sleep on Sleep Data: Influence of Sleep Physiological Signals on Stress Detection.

Soundarya Ramesh, Takahiro Masuda, **Hyung Woon Lee**, Yongquan Hu, Suranga Nanayakkara
NeurIPS - TS4H Workshop 2025

Bridging the Gulf of Endorsement: Enabling Automatic Endorsement of Cognitive Augmentation Systems.

Hyung Woon Lee, Dinithi Dissanayake, Dixon Prem Daniel Rajendran, Punyawee Anunpattana, Suranga Nanayakkara
(Under Review)

Sensing and Responding to Internal User States: A Systematic Literature Review of User-Aware Adaptive Assistive Wearables.

Dinithi Dissanayake, **Hyung Woon Lee**, Moritz Alexander Messerschmidt, Yize Wei, Suranga Nanayakkara
(Under Review)

RESEARCH & WORK EXPERIENCE

Augmented Human lab - National University of Singapore	Jan 2023 — Present
<i>Research Engineer (Full-time)</i>	<i>Sep 2025 — Present</i>
<i>Research Assistant</i>	<i>Jan 2023 — Jun 2025</i>

- Authored/co-authored several publications including NeurIPS 2025 workshop paper and under-review submissions. Initiated multiple projects on the theme of cognitive augmentation. Designed a multimodal encoder architecture for affect recognition. Developed the *EchoSense* mobile application for depression symptom tracking (OPPO Inspiration Challenge 2023 first round winner, 6.55% acceptance rate).

Centre for Holistic Inquiry into Lifelong Learning (CHILL) - National University of Singapore	Jul 2024 — Jun 2025
<i>Student Researcher</i>	

- Spearheaded development of integrated Learning and Engagement Measurement System (iLEMS) combining physiological signals and audiovisual data to measure student engagement. Implemented ProctorX, an AI-driven proctoring system using Computer Vision and MLLMs. Conducted final year project on multimodal stress detection using smartwatch sensors.
- **Awards:** Outstanding Undergraduate Researcher Prize (Individual)

Clinical Translational Sciences Lab (CTSL) - National University of Singapore	Jan 2023 — Jan 2025
<i>Research Assistant & Student Programmer</i>	

- Accelerated clinical study timelines by developing behavioral experiments using jsPsych and PsychoPy frameworks with integrated gaze tracking. Assisted in research report preparation and optimized data collection protocols.

Emplify (Startup)	Jan 2024 — Present
<i>Co-founder</i>	

- Spearheaded strategic planning and development of clinical tools using voice prosody and speech analysis for neuro-developmental disorder screening. Established healthcare partnerships and secured over \$100,000 SGD in grants through NUS Enterprise. Led product roadmap and business strategy for deployment of voice analysis tools and educational platform for professionals-in-training.

Robotic Assistant Dog for the Visually Impaired (Startup)	May 2022 — Jul 2023
<i>Research Engineer Intern</i>	

- Engineered cost-effective navigation system for Boston Dynamics Spot and Unitree Go1 robot dogs. Implemented SLAM algorithms and developed neural network pipeline integrating Mask2Former and Segment Anything (SAM) for collision avoidance using limited visual information.

TEACHING EXPERIENCE, EXTRACURRICULAR & VOLUNTEERING ACTIVITIES

National University of Singapore

Aug 2021 — Dec 2024

Senior Undergraduate Teaching Assistant

- Mentored 113 freshmen students for CS1101S: Programming Methodology across 4 semesters through intensive curriculum based on MIT's Structure and Interpretation of Computer Programs (SICP) textbook. Conducted weekly tutorial sessions on functional programming concepts, recursion, and computational thinking. Received teaching award nominations from 48 out of 113 students (42.5%). Teaching Feedback Rating: 4.7/5 (School Average: 4.2/5)
- Mentored 24 freshmen students for CS2030S: Programming Methodology II for 1 semester with weekly tutorial sessions. Served as an advisor and later mentor for CP2106: Independent Software Development Project.

Source Academy - National University of Singapore

Jul 2021 — Jun 2025

Senior Developer & Open Source Contributor

- Developed interactive features for Source Academy using ReactJS, focusing on game development modules and UI/UX design. Served on leadership team coordinating technical roadmap. Collaborated on research initiative analyzing student programming approaches through automated program classification.
- **Awards:** Outstanding Undergraduate Researcher Prize (Group)

MarkBind - National University of Singapore

Jul 2022 — Jun 2025

Senior Developer, Project Manager & Open Source Contributor

- Contributed feature enhancements and bug fixes to MarkBind, an open-source tool for generating static websites from markdown-like syntax. Participated in code reviews and collaborated with an international developer community on GitHub.

ACM UbiComp/ISWC 2025

Oct 2025

Student Volunteer

- Provided logistical support for conference sessions, workshops, and poster presentations. Assisted with registration, attendee coordination, and technical setup while facilitating networking opportunities between researchers and industry professionals.

Python Quest - NUS Centre for Computing for Social Good & Philanthropy (CCSGP)

Oct 2021 — Dec 2021

Volunteer Tutor

- Taught an introductory Python course to a group of 20 students diagnosed with Autism Spectrum Disorder (ASD) seeking to upskill.